

Statistical Mechanics Recitation Note

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DAY 1

- Microstate and Macrostate
- Essences of Binomial Distribution
- How Systems Reach Equilibrium State and How Equilibrium State Evolves in Time
- The Gaussian Approximation of Binomial Distribution

Microstate and Macrostate

Microstate: A microstate represents a specific instance or configuration of a system among all possible configurations. For a fair system, all microstates are equally likely to occur at any time.

Macrostate: A macrostate is defined as a set of microstates that share a common macroscopic property. Unlike microstates, macrostates are not equally probable, since different macrostates may encompass different numbers of microstates.

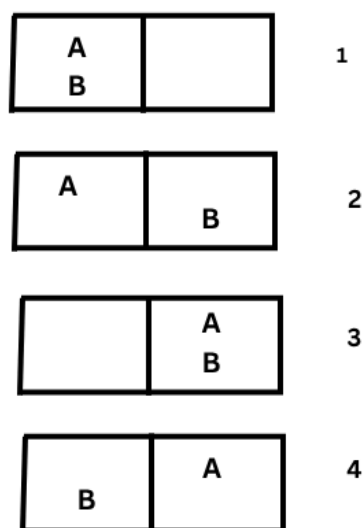


Figure1: Illustration of microstates and macrostates

for a simple system.

Example:

States 1, 2, 3, and 4 are microstates in figure .

States 2 and 4 correspond to a macrostate defined by the condition $n = 1$ and $n' = 1$ in figure .

Binomial Distribution

The binomial distribution arises from the probability of two mutually exclusive events, such as finding a particle in the left box or the right box.

- There are 2^N microstates for N binomial events.
- The macrostates depend on the probability of occurrence of each individual microstate and the total number of microstates that correspond to a particular outcome.
- The distribution is peaked at $n = N/2$ for $p = q$.

Equilibrium State

An **equilibrium state** is a state that is independent of time.

- Statistically, it corresponds to a macrostate that encompasses a significantly large number of microstates.
- Small fluctuations occur around this large macrostate.

Gaussian Approximation of Binomial Distribution

Near its maximum (which is around the mean), the variation of the binomial distribution is small compared to its amplitude. For large N , the binomial distribution can be approximated by a Gaussian distribution in this region.

(Refer to lecture notes, pages 16–17.)

Exercises

1. Consider rolling a fair die twice. Taking the sum of the two consecutive outcomes, list each microstate. Show that the microstate corresponding to the sum 7 is the most probable macrostate.

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2. Show that for a binomial distribution, the maximum of the distribution occurs at $n = N/2$ when $p = q = 0.5$. Then generalize the result for any p and q .

DAY 2

- Specification of State
- Statistical Ensemble
- Accessible States of a System
- Ergodic Hypothesis
- Reversible and Irreversible Processes

Specification of State

The state of a system (either microscopic or macroscopic) is described by certain measurable parameters.

Example:

- Classically, we use phase space with momentum and position coordinates.
- Quantum mechanically, the complete set of commuting operators of a system is used to describe the state.
- For a hydrogen atom, to fully specify its electron state, we use the quantum numbers n , l , m , and m_s .

Statistical Ensemble

A **statistical ensemble** is a very large set of possible states of a system that encompasses all of its microstates.

- In statistics, it literally represents a set of all possible event outcomes.

Accessible States of a System

The **accessible states** of a system are those states, constrained by specifying physical parameters. Leading to reduced total number of allowed microstates, that may not happen otherwise.

Example:

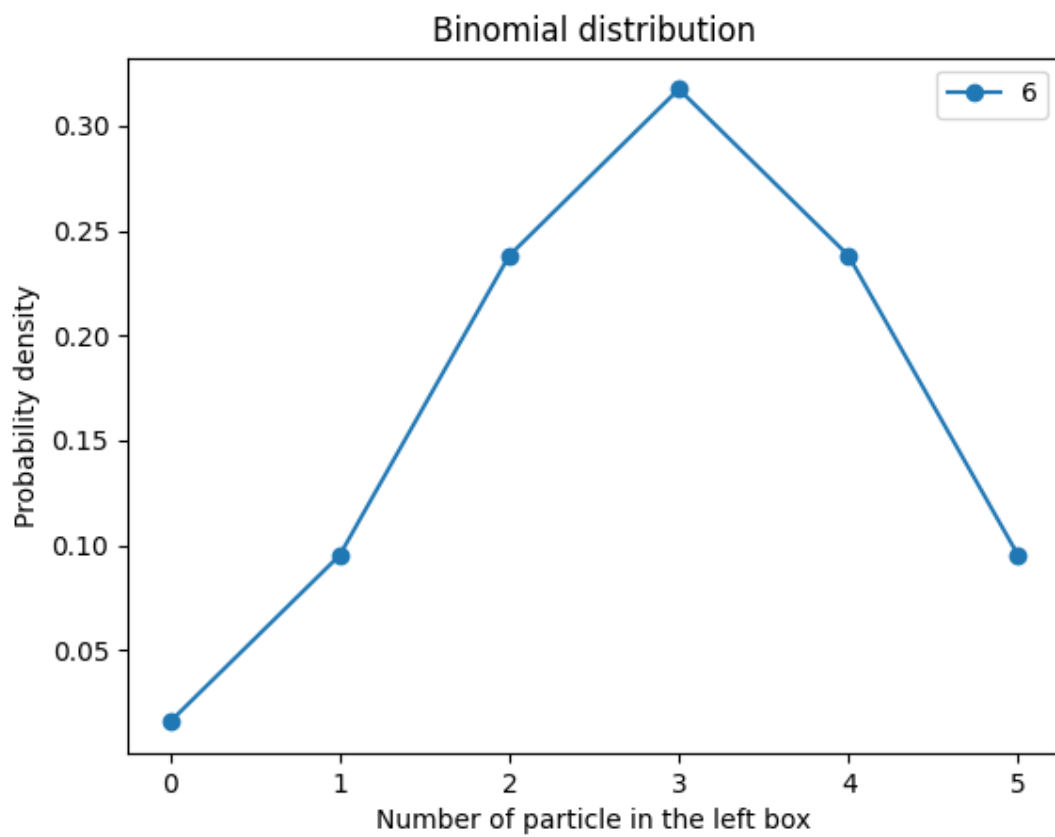


Figure 1: Binomial distribution with $N=6$ and $p=q=0.5$

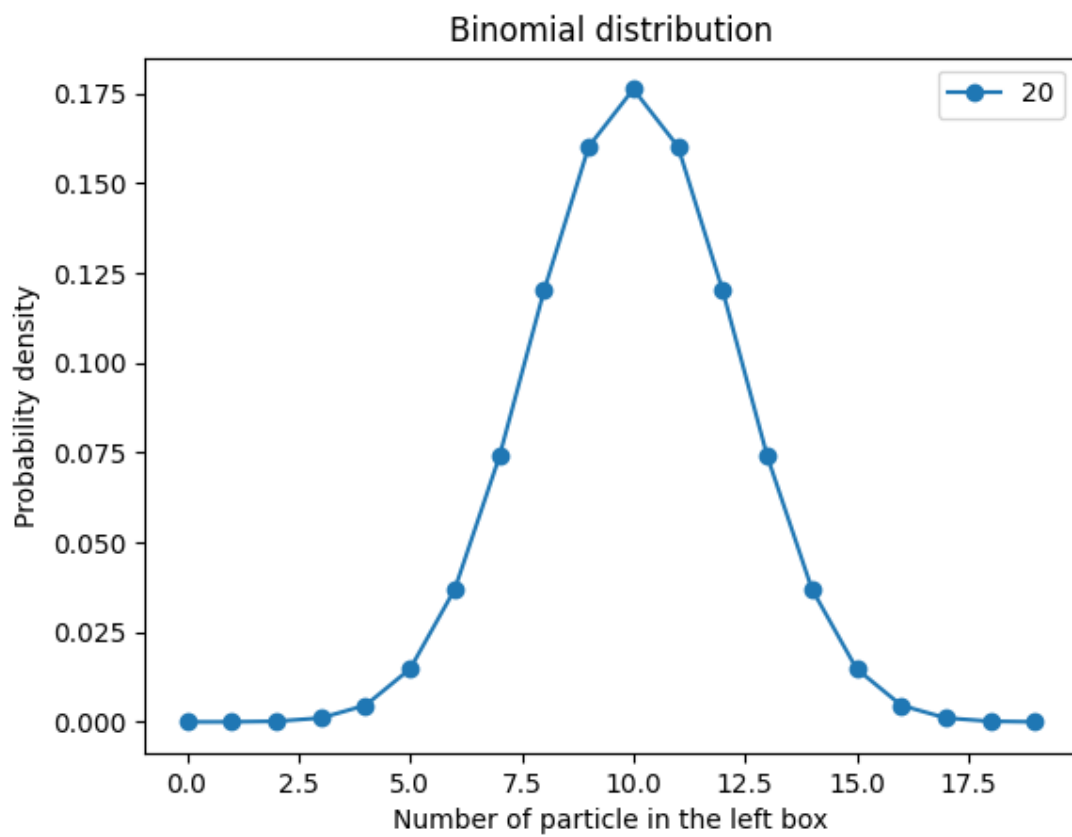


Figure 2: Binomial distribution with $N=20$ and $p=q=0.5$

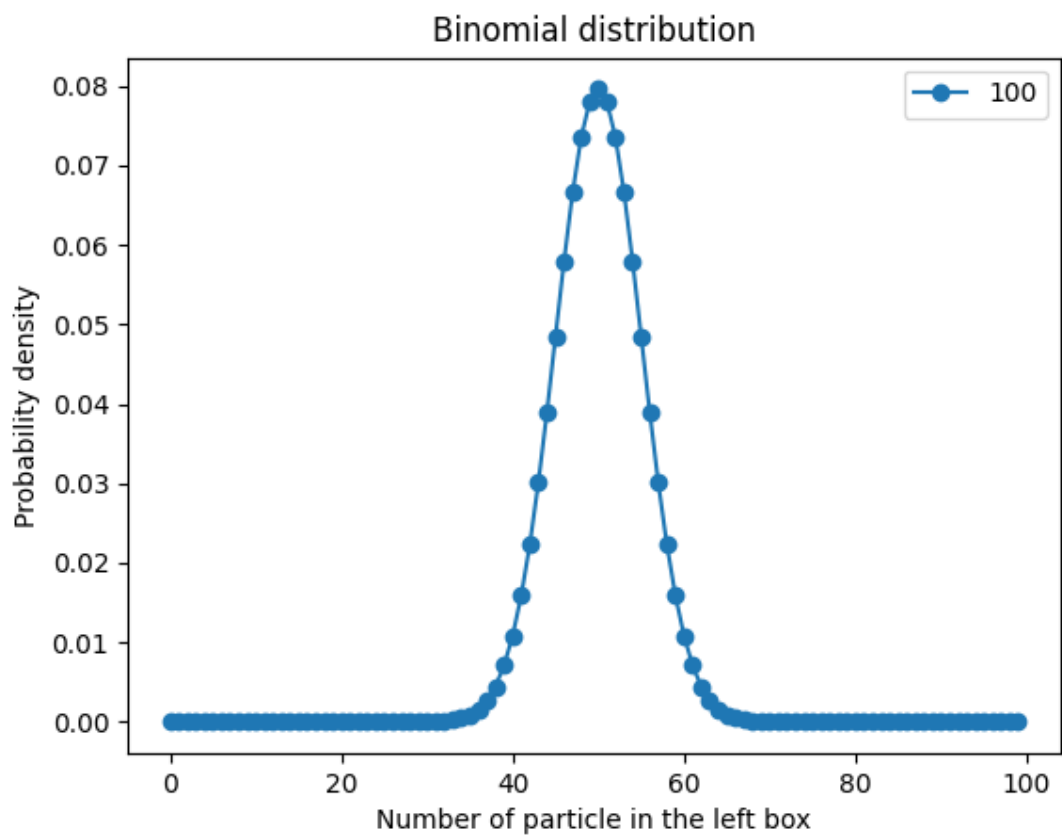


Figure 3: Binomial distribution with $N=10$ and $p=q=0.5$

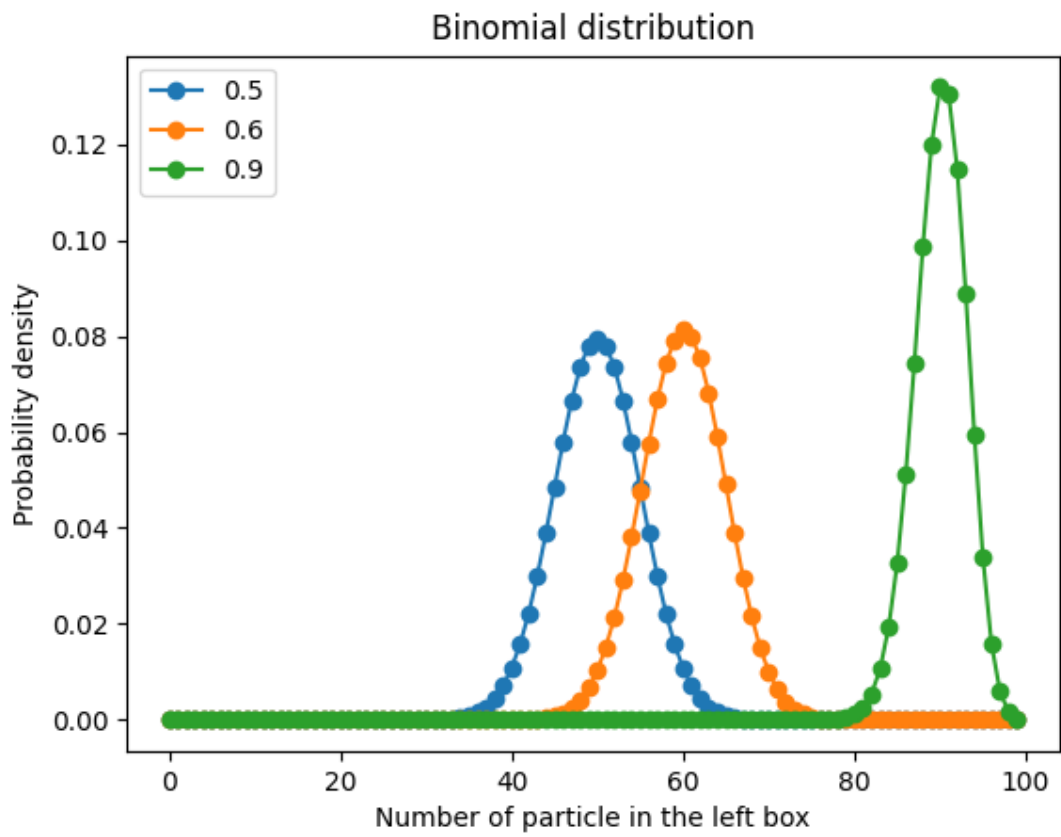


Figure 4: Binomial distribution with $N=100$ and $p=0.5$, $p=0.6$ and $p=0.9$

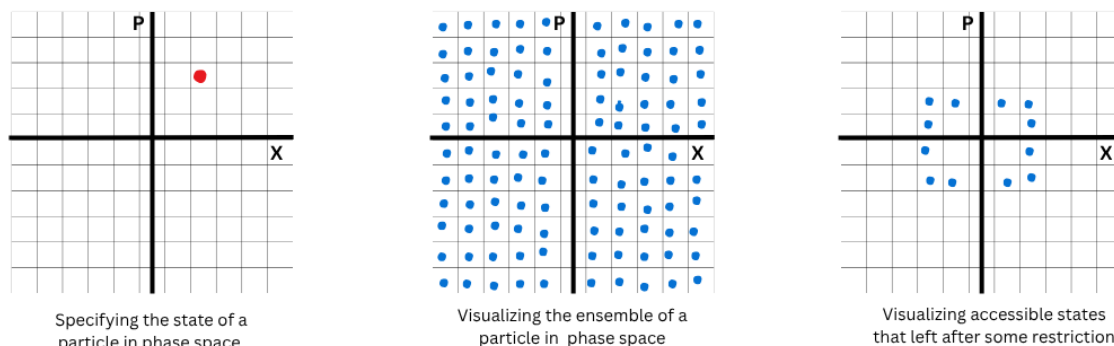


Figure 5: A depiction of states of a particle in phase. This is not to give precise image but make the ideas comprehensible

- Consider an isolated system of 5 particles. If the total system energy is E , then the sum of the individual particles energy must be equal to E . This constraint limits the number of ways the particles can share the energy.
- For a hydrogen atom in the second excited state ($n = 3$), there are 18 degenerate states (microstates). However, if the angular momentum state is specified say $l = 1$, the allowed degenerate states lowered to are 3.

Ergodic Hypothesis

The **ergodic hypothesis** states that, given enough time, a system will trace out all of its accessible microstates, spending an equal amount of time in each of them.

Reversible and Irreversible Processes

For an **isolated system**, if a perturbed (disturbed) system returns to its initial state, the process is said to be **reversible**. If it does not return to its initial state, the process is **irreversible**.

- A reversible process does not affect the ensemble; the system returns to its initial configuration.

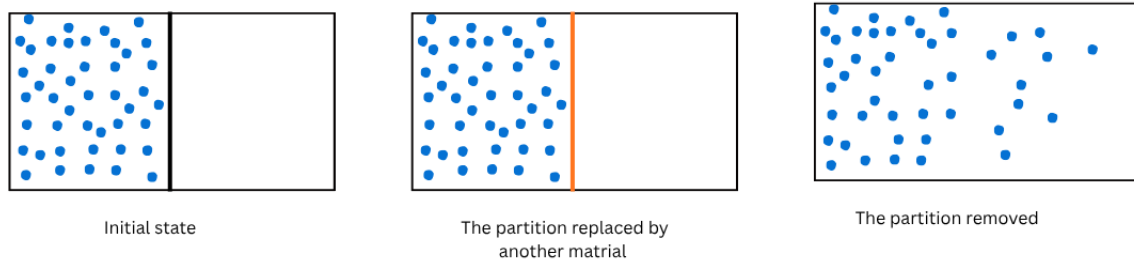


Figure 6: An illustration of reversible and irreversible processes; a) initial state, b) reversible process, c) irreversible process

- An irreversible process results in additional states to the ensemble, making it less probable for the system to return to its initial state.

Day 3

- Density of State
- Thermal Interaction (Definition of Temperature)
- Canonical Ensemble

Density of State

Suppose you measure the energy of a system to be E within a continuous range from E to $E + \delta E$. The **density of states** is defined as the number of microstates corresponding to this energy range, per energy width δE ¹. Hence, the term “density of microstates” is more precise.

Illustration: Consider a non-uniform rod as shown in figure 7. Over a very small length interval δx , its linear density may be assumed uniform with $\mu(x)$. Thus the mass of this segment of the rod is

$$\delta m = \mu(x) \delta x. \quad (1)$$

Similarly, for a system with energy uncertainty δE , assuming the microstates are uniformly distributed in this interval with density $\omega(E)$, the total number of microstates at energy E is

¹This is usually the case in real measurements: since one cannot measure energy with infinite accuracy, the best one can do is measure an energy E with some uncertainty δE .

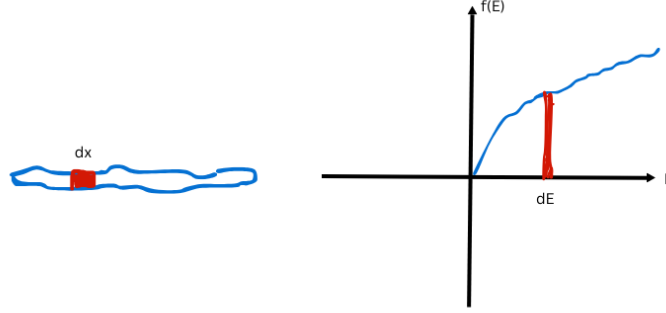


Figure 7: Illustrations to expose the analogous relation of mass density and density of state.

$$\Omega(E) = \omega(E) \delta E. \quad (2)$$

Mathematically, let $\phi(E)$ represent the number of microstates with energies less than E . Then,

$$\Omega(E) = \Phi(E + \delta E) - \Phi(E) \approx \frac{d\Phi}{dE} \delta E. \quad (3)$$

If we have f degrees of freedom with each $\phi(\epsilon)$ number of states for energy less than or equal to ϵ , whence $\epsilon \approx \frac{E}{f}$. Now we have those relations;

$$\phi(\epsilon) \propto \epsilon \approx \frac{E}{f} \quad (4)$$

$$\Phi(E) \propto (\phi(\epsilon))^f \quad (5)$$

Introducing this to equation 3, we reach to this ²

$$\ln \Omega(E) \approx f \ln \phi(\epsilon) \quad (6)$$

Thermal Interaction

Imagine two systems A and A' that are allowed to thermally interact via heat exchange as in figure 8. Let their energies be E and E' , respectively. The combined system A_0 has total energy

$$E_0 = E + E' \quad (7)$$

²This should remind us something related to entropy from our thermal and fluid course.



Figure 8: Two thermally interacting sub systems. The whole system is signified by A_0

Let the number of microstates available to each system at these energies be

$$\Omega(E), \quad \Omega'(E'), \quad \Omega_0(E_0),$$

for systems A , A' , and A_0 , respectively.

Since the systems are independent aside from energy exchange, the number of microstates crossponding to system A_0 when smaller system A contains energy E , could be given as

$$\Omega_0(E) = \Omega(E)\Omega'(E') \quad (8)$$

However, $\Omega_0(E)$ is proportional to the probability of finding a system A with energy E . Hence, we can ask what requirement should be fulfilled for the maximum probability state with energy E .

$$P(E) \propto \Omega_0(E) \quad (9)$$

$$\frac{d \ln P(E)}{dE} = \frac{d \ln \Omega(E)}{dE} + \frac{d \ln \Omega'(E')}{dE} = 0 \quad (10)$$

Recalling the conservation of energy from equation 7, $dE = -dE'$. Thus we arrive this fundamental relation

$$\frac{d \ln \Omega(E)}{dE} = \frac{d \ln \Omega'(E')}{dE'} = \frac{1}{\beta} \quad (11)$$

Definition of Temperature

We define a parameter

$$\beta = \frac{1}{kT},$$

where k is the Boltzmann constant and T is the temperature.

The temperature T characterizes whether or not two systems will exchange heat when placed in thermal contact. Systems with equal temperatures do not interact thermally.

Canonical Ensemble

A **canonical ensemble** is an ensemble of systems that are each allowed to thermally interact with a heat reservoir. Each system can exchange energy with the reservoir, but the temperature of the ensemble is fixed at the reservoir temperature.

Derivation of the Boltzmann Distribution

Suppose we have a small subsystem with energy ϵ and a large heat reservoir with energy $E - \epsilon$, as shown in Figure 9.

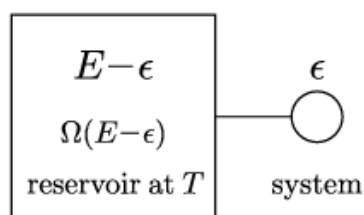


Figure 9: Subsystem of energy ϵ in thermal contact with a reservoir of energy $E - \epsilon$.

The probability that the subsystem has energy ϵ is proportional to the number of microstates of the reservoir:

$$P(\epsilon) \propto \Omega(E - \epsilon).$$

Expand $\ln \Omega(E - \epsilon)$ in a Taylor series around E :

$$\ln \Omega(E - \epsilon) = \ln \Omega(E) - \epsilon \left(\frac{d}{dE} \ln \Omega(E) \right) + \dots$$

Since $\epsilon \ll E$, higher-order terms are neglected. The derivative satisfies

$$\frac{d}{dE} \ln \Omega(E) = \beta,$$

where $\beta = \frac{1}{k_B T}$.

Thus,

$$\ln \Omega(E - \epsilon) \approx \ln \Omega(E) - \beta \epsilon.$$

Exponentiating:

$$\Omega(E - \epsilon) \propto e^{-\beta \epsilon}.$$

Therefore,

$$P(\epsilon) = C e^{-\beta \epsilon},$$

where C is a normalization constant.

If the subsystem has discrete energy levels ϵ_k , then

$$P(\epsilon_k) = \frac{e^{-\beta\epsilon_k}}{Z}, \quad Z = \sum_k e^{-\beta\epsilon_k},$$

which is the Boltzmann distribution.

Illustration

Assume a 10×100 grid box where each site initially contains one quantum of energy. When this grid interacts with a reservoir and exchanges one quantum at a time (only one quanta energy transfer is allowed), the number of sites with energies $0, 1, 2, 3, \dots, 300$ will evolve according to the Boltzmann distribution.

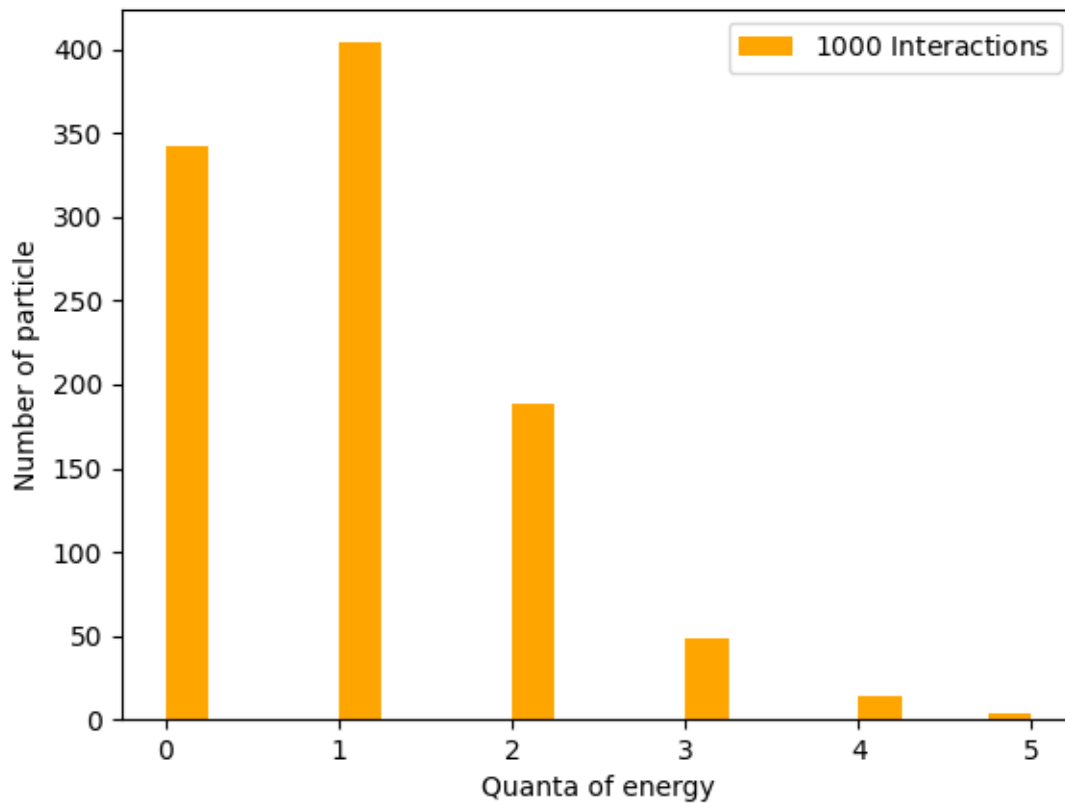


Figure 10: The number of quanta of energy distribution after a random of 1000 interactions

Day 5

- Quadratic approximation of potential energy (potential wells)

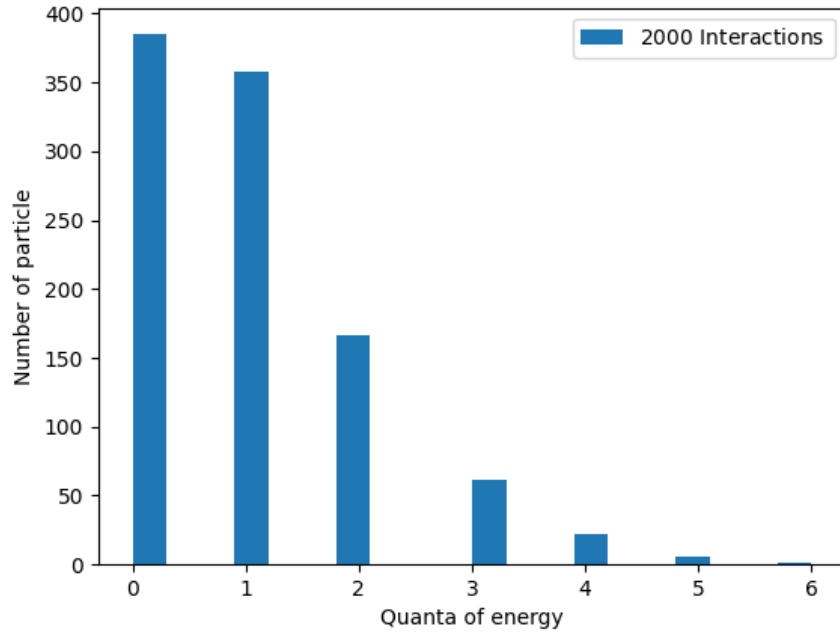


Figure 11: The number of quanta of energy distribution after a random of 2000 interactions

- The theorem of equipartition
- Applications of equipartition theorem
- Limitations of equipartition theorem

Potential Wells

Most potential energy functions are expandable about their minimum, giving a non-vanishing second-order term of the Taylor series:

$$U(x) = U_0 + \frac{1}{2} \left. \frac{d^2U}{dx^2} \right|_{x=0} x^2 + \mathcal{O}(x^3).$$

Hence, if one can find $U(x)$ in a thermally interacting system, the application is vastly transferable.

The Theorem of Equipartition

Assume a system thermally interacting with a heat reservoir, with total energy given by

$$E = \varepsilon_i(x_i) + E(x_1, x_2, \dots, x_n),$$

where ε_i is independent of external parameters but depends on x_i .

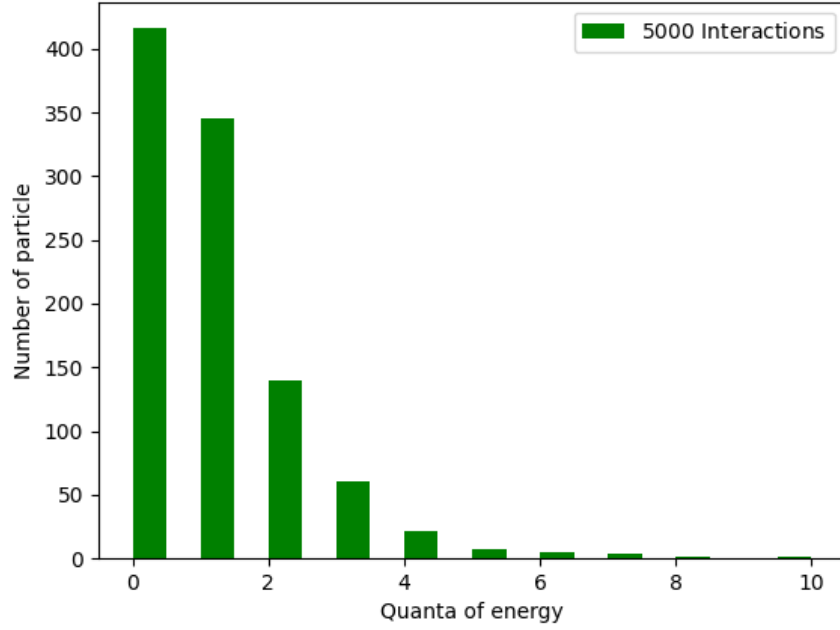


Figure 12: The number of quanta of energy distribution after a random of 5000 interactions

Assume

$$\varepsilon_i(x_i) = ax_i^2,$$

i.e., ε_i exhibits a quadratic relation with x_i .

Now,

$$\langle \varepsilon_i \rangle = \frac{\int \varepsilon_i e^{-\beta E(x_1, x_2, \dots, x_n)} dx_1 \cdots dx_n}{\int e^{-\beta E(x_1, x_2, \dots, x_n)} dx_1 \cdots dx_n}.$$

Separating the x_i -dependent part,

$$\langle \varepsilon_i \rangle = \frac{\int ax_i^2 e^{-\beta(ax_i^2 + E')} dx_i}{\int e^{-\beta(ax_i^2 + E')} dx_i},$$

where E' is independent of x_i .

Thus,

$$\langle \varepsilon_i \rangle = -\frac{1}{2} \frac{\partial}{\partial \beta} \ln \left(\int_{-\infty}^{\infty} e^{-\beta ax_i^2} dx_i \right).$$

Evaluating the Gaussian integral,

$$\int_{-\infty}^{\infty} e^{-\beta ax_i^2} dx_i = \sqrt{\frac{\pi}{\beta a}},$$

we obtain

$$\langle \varepsilon_i \rangle = \frac{1}{2\beta} = \frac{1}{2}kT.$$

This holds for any energy function with quadratic dependence on external parameters. Thus, for f degrees of freedom with quadratic interaction,

$$\langle E \rangle = \frac{f}{2}kT.$$

Applications of Equipartition Theorem

1. Harmonic Oscillator

Suppose you have a harmonic oscillator at room temperature. Its energy is given by

$$E = \frac{1}{2}mv^2 + \frac{1}{2}m\omega^2x^2, \quad \omega = \sqrt{\frac{k}{m}}.$$

Thus, the mean energy from the reservoir is

$$\langle E \rangle = 2 \left(\frac{1}{2}kT \right) = kT \approx 10^{-2} \text{ eV}.$$

This is therefore not noticeable on macroscopic scales.

2. Rotational Kinetic Energy

A diatomic molecule rotates about two axes. The rotational energy is

$$E_R = \frac{L_x^2}{2I} + \frac{L_y^2}{2I}.$$

Thus,

$$\langle E_R \rangle = kT.$$

3. Diatomic Molecule

A diatomic molecule has 7 degrees of freedom with quadratic energy dependence (3 translational, 2 rotational, and 2 vibrational):

$$E = \frac{1}{2}m(v_x^2 + v_y^2 + v_z^2) + \frac{1}{2}m\omega^2(x_1^2 + x_2^2) + \frac{L_x^2}{2I} + \frac{L_y^2}{2I}.$$

Hence,

$$\langle E \rangle = \frac{7}{2}kT.$$

4. Brownian Motion

Biological objects such as pollen grains appear to be in continuous motion when placed in a fluid. This is due to kinetic energy borrowed from the reservoir.

$$\langle E \rangle = \frac{1}{2}m\langle v_x^2 \rangle = \frac{1}{2}kT \quad \Rightarrow \quad \langle v_x^2 \rangle = \frac{kT}{m}.$$

Limitations of Equipartition Theorem

- Often energy is quantized; thus, the result breaks down in the low-temperature limit.
- For potential wells, the energy should not be dominated by anharmonic terms; hence the temperature must not be too high.

Remark: The equipartition theorem applies at sufficiently high temperatures to ignore quantization and sufficiently low temperatures for the quadratic approximation to remain valid.

Exercises

1. Consider a system in thermal equilibrium whose energy is given by

$$E = b x,$$

where b is a constant and x is a continuous parameter. Show that the thermal average of the energy is

$$\langle E \rangle = kT.$$

2. Using the result of part (1), show that the difference between the molar heat capacities at constant pressure and constant volume satisfies

$$C_P - C_V = R.$$